QR SHIELD

System Architecture • CYBER BOYS

1. Purpose

QR Shield is a browser-based prototype that helps a user inspect decoded QR content before deciding whether to open a destination. It provides explainable, rule-based indicators rather than claiming to prove that a QR code is safe.

2. System architecture

INPUT LAYER
Camera scan • QR image upload • pasted content

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QR DECODING
html5-qrcode library (browser/device support required)

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LOCAL ANALYSIS ENGINE
JavaScript rules • URL normalization • payload-type detection • suspicious-pattern checks

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RESULT & GUIDANCE
Heuristic SAFE / CAUTION / RISK label • findings • guidance • high-risk sound alert

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LOCAL RECORDS & EXPORT
Browser-stored scan history • duplicate detection • CSV/JSON export • report utilities

3. Technology stack

Frontend: HTML, CSS, JavaScript. QR decoding: html5-qrcode. QR generation: qrcodejs. Browser Web APIs support local storage, sound, copying, and file export where permitted.

4. Data flow

1) User scans a QR code, uploads a QR image, or pastes decoded content. 2) The decoder returns the payload. 3) JavaScript classifies the payload type and applies local URL/pattern checks. 4) The app shows a heuristic label and findings. 5) The user may review history or export available data.

5. Trust boundaries and limitations

The supplied presentation describes a browser-side prototype with no backend and no live AI service. It does not perform live reputation lookups or final redirect inspection. A SAFE label is not proof of safety, and the heuristic

score is not a fraud probability. Browser storage is not a secure account system; external CDN libraries require network access.

6. Recommended next engineering steps

Add automated unit tests and a curated test set; measure false positives and false negatives; add a carefully designed redirect-inspection service only after security review; document data retention; and test camera permissions and accessibility on target devices.

Prepared from the supplied QR Shield presentation. Architecture describes the prototype as presented and does not independently verify every feature in the running code.